

Output:

-- Binary Search Tree Menu --

1. Insert Into BST
2. Inorder Traversal
3. Preorder Traversal
4. Postorder Traversal
5. Display BST
6. Exit

Enter choice : 1

Enter value to insert : 40

Enter choice : 1

Enter value to insert : 25

Enter choice : 1

Enter value to insert : 60

Enter choice : 2

Inorder Traversal: 25 40 60

Enter choice : 3

Preorder Traversal: 40 25 60

Enter choice : 4

Postorder Traversal: 25 60 40

Enter choice : 5

BST element (Inorder): 25 40 60

Enter choice : 6

22/12/2021

Q1) Write a program to traverse a graph using BFS method.

Pseudocode:

Start

read number of vertices

Read adjacency matrix

set visited[i] = 0 for i = 0 to n-1

read starting vertex

create empty queue, initialise front and rear = 0

visited[start] = 1, add start into the queue

for i = 0 to n-1

if graph[node][i] = 1 and visited[i] == 0



visited[i] = 1

insert the value of i into queue

stop

code:

```
#include <stdio.h>
```

```
int graph[20][20], visited[20], n;
```

```
void BFS(int start) {
```

```
    int queue[20], front=0, rear=0;
```

```
    visited[start] = 1;
```

```
    queue[rear++] = start;
```

```
    while (front < rear) {
```

```
        int node = queue[front++];
```

```
        printf("%d", node);
```

```
        for (int i=0; i<n; i++) {
```

```
            if (graph[node][i] == 1 && !visited[i]) {
```

```
                visited[i] = 1;
```

```
                queue[rear++] = i;
```

```
            }
```

```
        }
```

```
    }
```

```
}
```

```
int main() {
```

```
    int start;
```

```
    printf("Enter number of vertices:");
```

```
    scanf("%d", &n);
```

```
    printf("Enter adjacency matrix:\n");
```

```
    for (int i=0; i<n; i++) {
```

```
        for (int j=0; j<n; j++) {
```

```
            scanf("%d", &graph[i][j]);
```

```
        }
```

```
    }
```

```
    printf("Enter starting vertex:");
```

```
    scanf("%d", &start);
```

```
    printf("BFS Traversal:");
```

```
    BFS(start);
```

```
    return 0;
```

```
}
```


output:

Enter number of vertices: 4

Enter adjacency matrix:

0	1	1	0
1	0	0	1
1	0	0	1
0	1	1	0

Enter starting vertex: 2

BFS Traversal: 2 0 3 1

pi 22/12/21